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Formal Language & Finite Automata

Turing machines

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Turing machines - examples

Construct Turing machines recognizing the following languages.

1. $\{w\#w \mid w \in \{0,1\}^*\}$
2. $\{a^n b^n c^n \mid n \geq 0\}$
3. $\{0^{2^n} \mid n \geq 0\}$

Recursive enumerable and recursive languages

Given a Turing machine M and an input string w there are three outcomes.

1. M accepts w ;
2. M rejects w ; or
3. M never halts (enters an infinite loop).

Recognizing vs. deciding

Definition

A language L is **recursive enumerable** or **Turing recognizable** if it is recognized by some Turing machine, that is there is a Turing machine M such that $L(M) = L$.

Definition

A language L is **recursive** or **Turing decidable** if it is decided by some Turing machine, that is there is a Turing machine M such that $L(M) = L$ and M halts on every input string.

We have seen that $\{w\#w \mid w \in \{0,1\}^*\}$, $\{a^n b^n c^n \mid n \geq 0\}$ and $\{0^{2^n} \mid n \geq 0\}$ are Turing decidable.